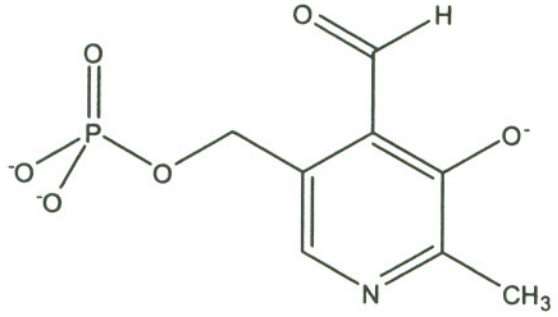
Biochemistry Cumulative Examination

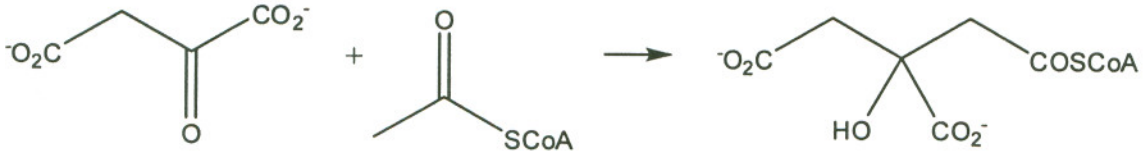
January 22, 2010

Answer the following questions about enzymatic mechanisms.

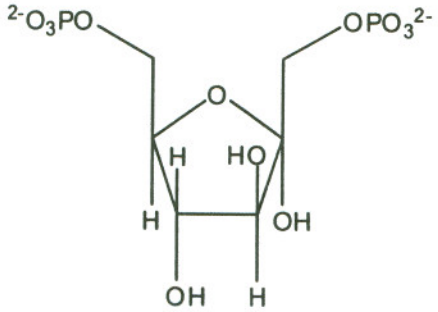
1. Alanine aminotransferase catalyzes the transfer of the amine of alanine to α-ketoglutarate yielding pyruvate and glutamate. It, of course, catalyzes the reverse reaction as well. The prothestic group pyridoxal phosphate, shown below, plays the key role in this transfer. Give a mechanism (arrow pushing, intermediates, etc.) that shows how pyridoxal phosphate carries out this conversion. 30 points.



2. Citrate synthase adds acetyl CoA to oxaloacetate without using any prosthetic groups. As shown below, the initial product of this reaction is citryl CoA. Show a mechanism for this reaction. 30 points



3. Aldolase cleaves fructose 1,6-bisphosphate below into dihydroxyacetone phosphate and glyceraldehyde 3-phosphate, again without using any prosthetic groups. Show a mechanism for this reaction. 30 points



4. Triose phosphate isomerase very efficiently catalyzes the conversion of dihydroxyacetone phosphate to glyceraldehyde 3-phosphate (below). Show a mechanism. 10 points.